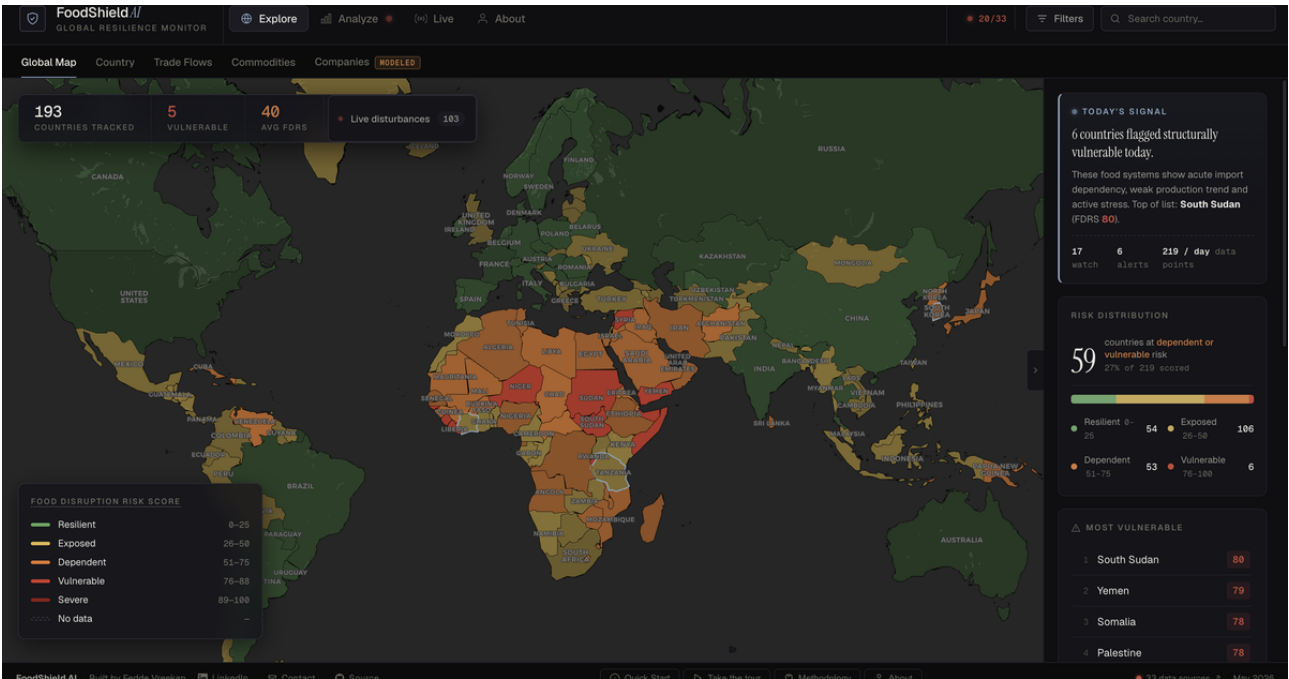


FoodShield AI

An open dashboard scoring food-supply fragility for 193 countries and 50 US states from 33 public data pipelines, refreshed every 6 hours.



FoodShield AI global view — Food Disruption Risk Score (FDRS) coloured by tier. Live at foodshield-ai-fv.vercel.app

Built by	Fedde Vreeken — International Economics & Business Economics, Erasmus School of Economics
Live URL	https://foodshield-ai-fv.vercel.app
Source code	https://github.com/feddevreeken-ctrl/foodshield-ai
Data sources	33 public pipelines (FAO, World Bank, USDA, UN Comtrade, ACLED, WFP, FEWS NET, NAS)
Refresh cadence	Every 6 hours via GitHub Actions (00 / 06 / 12 / 18 UTC)
Coverage	193 UN-member countries + 50 US states (~258 territorial entries total)
Output	Food Disruption Risk Score (FDRS), 0–100, per country + per US state
License	Code MIT · Data attributed to original providers

Executive summary

FoodShield AI is an open, free, no-account-required dashboard that estimates how exposed each country in the world is to food-supply disruption — and explains why. It combines trade dependency, climate exposure, conflict, logistics, supplier concentration, prices, and live disturbance feeds into a single composite indicator — the **Food Disruption Risk Score (FDRS)**, on a 0–100 scale where 0 is fully resilient and 100 is maximally vulnerable.

Every score on the site is auditable. Each component traces back to its underlying public dataset, each methodology decision is documented inline, and a per-source health manifest shows in real time which pipelines are live, degraded, or relying on manual snapshots. Where the site uses modelled values rather than observed data, it labels them as such.

The project was built solo over roughly four months, end to end: data engineering, scoring methodology, frontend, and deployment. It is not a product. It is a working research artefact that exists because the public infrastructure for tracking global food fragility is fragmented across dozens of agencies, formats, and update cycles — and there was no single place to see the whole picture at once.

Who it is for

Researchers, journalists, policy teams, supply-chain analysts, and students who want a fast, transparent, source-traceable view of global food risk. It is explicitly **not** calibrated for trade execution, humanitarian targeting, or sovereign-risk pricing — those are decision-grade uses that require a back-testing pipeline that has not yet been built.

Top-line numbers

Countries tracked	193 UN-member states + 50 US states + ~21 territories
Data pipelines	33 public sources, refreshed every 6 hours
FDRS components	7 weighted components (28 / 18 / 14 / 14 / 9 / 9 / 8)
Nowcast signals	15 live sub-signals bounded to [–10, +35] points
Tier classification	5 bands: Resilient · Exposed · Dependent · Vulnerable · Severe
Code size	~22,750 lines (single-file HTML + 33 Python refresh scripts)

What the site does

The site is organised around four navigation groups — **Explore, Analyze, Live, About** — each containing the visualisations and tools below.

Explore

Global Map. A 2D world map coloured by FDRS tier (Resilient → Severe). Hover any country for its score; click for a full country panel with score breakdown, trend chart, import/export composition, top 3 nearest peer countries, and an AI-generated narrative summary of the country's current food-security situation.

Country. A searchable browser of all 193 countries with structural and nowcast FDRS side by side, plus the seven component scores expanded individually.

Trade Flows. The Trade Flow Atlas — an interactive globe view of bilateral staples trade across 10 HS commodity codes (wheat, rice, maize, soybeans, palm oil, sugar, coffee, cocoa, fertilizer, beef). Lines show real Comtrade-observed flows; modelled routes for 27 commodities × 436 supplier-mix combinations fill gaps where Comtrade is thin.

Commodities. Per-commodity detail pages with World Bank Pink Sheet price history (60-month sparkline), top producers, top exporters, supplier-concentration ranking, and exposure heatmap.

Companies. Per-trader (Cargill, ADM, Bunge, COFCO, Wilmar, Olam, JBS, Tyson, LDC, Nutrien, Viterro, Yara) sourcing-footprint cards. Labelled **MODELED** — no trader publishes per-country sourcing volumes, so these are directional risk-screens, not quantitative supply-chain weights.

Analyze

Score. Global rankings, country comparison, and a 2×2 dependency matrix (caloric import dependency × supplier concentration).

Scenario stress-test. A what-if engine. Users select shocks — major wheat-supplier export shock, rice export ban, maize tariff, fertilizer chokepoint, drought, climate shift, conflict in a chokepoint region — and the engine recomputes FDRS for every country. Uses a channel-overlap damper ($\sqrt{\sum \text{kick}^2}$ instead of straight sum when 2+ shocks hit the same FDRS component) to avoid double-counting. Provenance-aware: shocks applied to sourced data carry full impact; shocks applied to legacy curated data are damped to 60%.

Modeled outlook to 2030. A medium-term projection per country combining the structural baseline (heritage curated values) with a modelled live overlay. Explicitly labelled *illustrative scenario estimate, not an official FAO-OECD or CCKP forecast*.

Live

Live disturbances. Time-ordered feed of conflict events (ACLED), humanitarian alerts (ReliefWeb), natural-disaster alerts (GDACS), flood discharge anomalies (Open-Meteo Flood + USGS Water), weather extremes (Open-Meteo), active fires over cropland (NASA FIRMS), and air-quality stress (OpenAQ). Each event has a verified provenance pill (LIVE vs BASELINE).

About

Methodology. Inline documentation of the FDRS formula, sub-signal definitions, nowcast adjustment math, version-changelog, worked examples (Egypt, Sudan, Yemen).

Data Status. Per-source audit page: 33-row table showing last refresh timestamp, health status, what each feed powers downstream, and which pipelines are currently degraded with reason. Status pill in the global header summarises the manifest (20/33 healthy at time of writing).

How it was built

FoodShield AI is intentionally a single-file static web app on the front end, paired with a Python data layer that runs on GitHub Actions and writes JSON snapshots back into the repository. There is no server, no database, and no auth. The frontend reads the JSON files Vercel serves alongside the HTML.

Stack at a glance

Frontend	Single-file HTML (~22,750 lines) with vanilla JavaScript, D3.js for the map and trade-flow globe, C
Data layer	33 Python scripts under scripts/, each handling one upstream API or bulk download. Shared helper
Orchestration	GitHub Actions workflow (.github/workflows/refresh-data.yml) runs every 6 hours, executes scripts
Hosting	Vercel static deployment with custom vercel.json routing for /data/* and image assets
Auth-required APIs	ACLED, FEWS NET, UN Comtrade Plus, OpenAQ, NASA FIRMS, ReliefWeb — credentials in Git
Version control	Git, public on GitHub (~120 tracked files, 16+ active commits per week from auto-refresh)

Why a monolith?

The site is deliberately a single HTML file rather than a React or Next.js application. Reasons: (1) zero build step — the file Vercel serves is literally the file in the repo; (2) view-source is the documentation — anyone can read the rendering logic; (3) no client-side state-management complexity for a fundamentally read-only dataset; (4) page-load is one network round-trip plus the JSON snapshots, which makes it fast even on constrained connections (relevant for users in the regions the dashboard most concerns). The monolith will eventually be split into per-tab modules; that's on the post-launch backlog, not a launch blocker.

Data flow

Every 6 hours, the GitHub Actions workflow:

- 1. Checks out the repository and installs Python dependencies from scripts/requirements.txt.
- 2. Runs scripts/run_all.py, which dispatches each refresh_*.py in parallel where possible.
- 3. Each refresh script fetches its upstream (with retry + backoff), normalises to a stable JSON schema with per-field provenance, and writes to data/.
- 4. scripts/build_nowcast.py merges 15 live signals into data/nowcast.json — the per-country delta added on top of the structural FDRS.
- 5. scripts/build_source_manifest.py inspects every snapshot, derives status (ok / manual / degraded / setup_required / failed), and writes data/source_manifest.json.
- 6. scripts/validate_data_integrity.py runs assertion checks (≥150 countries, required ISO codes, no FAS-code swaps); workflow hard-fails if violated.
- 7. Workflow commits the diff back to main; Vercel auto-deploys.

Provenance discipline

Every value on the site carries one of five canonical quality flags. This is the spine of the methodology — without it, no number could be defended.

SOURCED	Fetches live from a public pipeline. Green pill. Refreshable on demand.
MANUAL	Hand-maintained snapshot of an official release (e.g. Feeding America's Map the Meal Gap, published a
MODELED	Computed at render time from sourced inputs (e.g. Supply-Chain Exposure, supplier shares for commod
CURATED	Legacy hand-authored heritage value awaiting re-derivation from a sourced pipeline. Used as fallback wh
ILLUSTRATIVE	Plausible example value with no upstream data anchor (e.g. trader sourcing footprints — no company pu

How I actually built it — a multi-AI workflow

FoodShield AI was built by one person, but not alone. The build sprint relied heavily on a rotation of AI tools, each chosen for the kind of work it does best. This section documents that workflow honestly — both because the project would not exist in its current scope without it, and because the methodology of *how* the dashboard was constructed is itself relevant to anyone trying to assess what the dashboard is.

The starting point

The first version was a single static HTML file with hand-coded country data and a Mapbox layer, built over a weekend in early 2026 as a coursework side-project on global trade dependency. It became FoodShield AI when I decided to extend it from a snapshot-of-2024 visualisation into a live, source-traceable system. That extension is what required the AI tooling — fetching, parsing, and validating 33 public data sources, each with its own schema quirks, was beyond what I could write by hand in the time I had.

The AI tools I used and what each one did

Claude Code (Anthropic). The primary build environment for most of the sprint. Used for: end-to-end feature work on `index.html`, scoring engine logic, the scenario stress-test channel-overlap damper, the trade-flow atlas rendering, the per-country panel UX, and the methodology page itself. Most of the audit-blocker bug fixes in the launch sprint were done in Claude Code sessions — including the USDA PSD FAS-code ISO swap bug (Belarus showing as Bolivia), the empty-string supplier-match bug that inflated Egypt's wheat exposure to \$15B, and the centroid-override system that stopped France rendering in Spain.

ChatGPT (OpenAI, GPT-4 / GPT-5 class). Used for second-opinion code review, alternative phrasings for the methodology copy, and as a sanity check on the FDRS weighting logic. When two AI assistants converge on the same answer to a question about indicator weighting or normalisation, that's a stronger signal than either of them individually. When they disagree, that's a flag to read the underlying paper myself rather than trust either.

Codex (OpenAI, via API). Used for batch refactoring across the 33 refresh scripts — particularly for standardising the `retry / backoff / provenance-tagging` boilerplate after the shared `_common.py` helpers were extracted. Codex's strength on purely mechanical Python transformations was the right tool for that pass.

Lovable. Used to prototype an alternative Next.js / React frontend in parallel with the monolith HTML, as a check on whether splitting the frontend would be a clear win or premature optimisation. The conclusion: the single-file HTML wins on simplicity and load speed for a fundamentally read-only dataset, and the React prototype is currently in the `foodshield/` folder of the repo as a reference for when the eventual split happens. Lovable also confirmed the v22 design direction (header layout, country panel sidebar collapse, methodology accordion) before those changes shipped.

Gemini (Google). Used as an adversarial reviewer at three checkpoints: pre-launch methodology audit, post-launch source-claim audit, and final copy honesty pass. Gemini's tendency to push back hard on overclaims was useful — it caught the original 'Forecast' naming, flagged the 'famine' miscount, and pushed for the explicit removal of confidence-interval UI that wasn't grounded in a back-test pipeline. Several of the most important honesty fixes in the launch sprint came from Gemini's pushback.

Groq (Llama hosting). Used for fast, low-stakes high-throughput tasks: bulk country-name normalisation across pipelines (ISO3 mismatches between FAO, USDA, and World Bank), bulk caloric-share sanity checks against FAO Food Balance Sheets, and translation of source-page descriptions where the original was non-English. Groq's speed advantage matters when you need to run 200 small lookups rather than 1 complex reasoning task.

The audit chain

The methodology page itself went through an explicit adversarial-review pipeline: Claude Code drafted, Gemini critiqued, ChatGPT proposed alternative framings, and I made the final call. Every major copy change in the launch sprint went through at least two of the four reviewers — not because any single one was unreliable, but because indicator framing is a high-stakes part of the project and three independent reads catch things one doesn't.

The external design audit done by a separate Anthropic-affiliated reviewer (a senior designer reviewing the site independently) flagged ten items pre-launch. I shipped three, skipped seven with documented rationale. The full audit log is in `CONVERSATION_LOG.md` in the repository. The decision to skip items like 'collapse 12 tabs into 5' was deliberate: the proposed simplification would have broken existing LinkedIn deep-links into the site and lost the path-picker conversion lift that the Quick Start modal generates.

Why I'm transparent about this

Two reasons. First, the AI-tool stack is part of what makes the project replicable — anyone with a similar dataset ambition and a domain they care about can now build comparable infrastructure with a small fraction of the engineering headcount that would have been required five years ago. The build cost was measured in API spend and weekends, not in person-quarters. That matters for the future of public-good data tooling.

Second: a dashboard built this way deserves more scrutiny, not less. Multi-AI workflows are excellent at producing fluent, plausible-sounding output. They are not excellent at knowing when they're wrong about something domain-specific. The provenance discipline described in section 03 — every value carries one of five quality flags, every component traces to its source, every modelled fallback is labelled — exists precisely because the build process required it. The site is auditable because it had to be.

What I wrote myself

The scoring methodology itself — which seven components, what weights, why those weights, where the boundaries between structural and nowcast fall — is mine. The AI tools executed implementations, suggested phrasings, and reviewed claims. The substantive decisions about *what to measure and how to weight it* came from reading the underlying literature (FAO methodological notes on food security indicators, IFPRI working papers on composite indicator construction, the FEWS NET IPC technical manual) and deciding what made sense for the use case. The AI tools cannot do that part. They can tell you what other people have done; they cannot tell you what to do.

The FDRS formula — full methodology

The Food Disruption Risk Score (FDRS) is a weighted composite of seven components, each normalised to 0–100 before weighting. The headline value shown in country panels is the **nowcast-adjusted** FDRS — the structural composite plus a bounded live-signal delta.

Structural formula

FDRS_{structural} = 0.28 · Import Dependency
+ 0.18 · Supplier Concentration
+ 0.14 · Production Trend
+ 0.14 · Food Inflation
+ 0.09 · Climate Vulnerability
+ 0.09 · Conflict / Logistics
+ 0.08 · Supply-Chain Exposure

Weights sum to 1.00. Output rounded to integer, clipped to [0, 100].

Component definitions

Weight	Component	What it measures	Primary source
28%	Import Dependency (caloric)	Share of national caloric supply imported	FAO Food Balance Sheets
18%	Supplier Concentration	Herfindahl-Hirschman Index of top-5 import partners	UN Comtrade + FAO
14%	Production Trend	5-year CAGR of domestic staple production	USDA PSD 2018-2024
14%	Food Inflation	Annual food CPI change (live cascade)	WFP > Eurostat > FAOSTAT > WB
9%	Climate Vulnerability	Composite: vulnerability, water stress, projected ΔTND-GAIN + Aqueduct + CCKP	
9%	Conflict / Logistics	Composite: humanitarian risk + rule of law + LPI	INFORM + WGI + LPI
8%	Supply-Chain Exposure	Trade-import dependency across staple commodities	FSB + Comtrade + PSD

Component-level methodology detail

Import Dependency (28%, caloric view). The share of a country's caloric supply that arrives via imports rather than domestic production. Sourced from FAO Food Balance Sheets. Normalised against the global distribution: 95%+ import-reliance maps to a score close to 100, fully self-sufficient countries map to ~0. This is explicitly the caloric view, not the trade-volume view — the Netherlands eats almost no rice domestically, so its caloric rice-dependency is ~0%, even though it imports ~\$481M of rice annually. The Supply-Chain Exposure component (8%) captures the trade-volume view separately so neither perspective is lost.

Supplier Concentration (18%). The Herfindahl-Hirschman Index of the top-5 import source countries for staple commodities. $HHI = \sum s_i^2$ where s_i is each supplier's share. Normalised so a single-source monopoly scores 100 and a perfectly diversified mix scores 0. This is the channel that flagged Sudan's exposure when Russia (40% of Sudan's wheat) halted exports — concentration matters even when total dependency is moderate.

Production Trend (14%). The 5-year compound annual growth rate of domestic staple production (wheat + rice + maize + soybeans), from USDA PSD 2018–2024. Negative trends score higher. Normalised against global peer distribution so a country with –2% CAGR in a world of +1% CAGR scores meaningfully worse than one with –2% in a generally declining world.

Food Inflation (14%). Annual food CPI change, read live from a four-tier source cascade: WFP HungerMap (for crisis countries with country-level coverage) → Eurostat HICP (EU monthly, CP011) → FAOSTAT food CPI (global annual) → World Bank headline CPI as a last-resort fallback. Calibrated so >60% annual food inflation scores ≈ 100 , near-zero scores ≈ 0 .

Climate Vulnerability (9%). When the underlying pipelines have country coverage, this is a 60% sourced / 40% heritage composite: ND-GAIN food-vulnerability sub-score $\times 0.45$ + Aqueduct drought risk $\times 0.20$ + Aqueduct baseline water stress $\times 0.15$ + CCKP projected temperature $\Delta \times 0.20$. CCKP and ND-GAIN are currently degraded (CCKP covers 0 countries in the served snapshot; ND-GAIN's most recent download failed) — the component falls back to the heritage curated value, flagged explicitly in the methodology page.

Conflict / Logistics (9%). 60% sourced / 40% heritage composite: INFORM Risk $\times 0.45$ + World Bank WGI Rule of Law $\times 0.25$ + World Bank Logistics Performance Index $\times 0.20$ + INFORM lack-of-coping-capacity $\times 0.10$. High humanitarian risk + weak rule of law + poor logistics scores close to 100. The LPI integration is post-v20.26 — earlier versions used a heritage-only conflict score.

Supply-Chain Exposure (8%, v20.2). Trade-volume weighted import dependency across staple commodities, premium-adjusted for supplier concentration. Captures sourcing risk that the caloric view misses. Derived from FAO Food Balance Sheets + UN Comtrade + USDA PSD. Resolves the Netherlands-rice class of issue described under Import Dependency.

Nowcast adjustment Δ

The nowcast FDRS adds a live-signal delta to the structural score, bounded to $[-10, +35]$. This bound exists so live noise can never overwhelm the structural baseline; a single bad month of inflation data cannot drive a structurally resilient country into the Severe tier on its own.

```
 $\Delta_{\text{nowcast}} = \text{ipc\_pressure (0-12)}$   
+ wfp_consumption (0-6)  
+ conflict (0-5)  
+ global_price (0-2)  
+ fx_shock (0-3) – 90d local-currency fall vs USD  
+ inflation_shock (0-3) – food inflation > 15%  
+ weather_extremes (0-4)  
+ flood (0-3) – river discharge anomaly  
+ fire (0-2) – NASA FIRMS over cropland  
+ air_quality (0-1) – OpenAQ PM2.5  
+ us_water (0-2) – USGS river-flow anomaly  
+ inform_amp (0-3) – when INFORM  $\geq 7.5$   
+ governance_drag (0-2) – when WGI rule-of-law  $\leq -1.0$   
+ psd_shortfall (0-3) – USDA PSD staples production gap  
– humanitarian_response (0-2) – active large-scale aid presence
```

Each signal is computed independently per country, capped at its individual maximum, then summed and clamped to the global ± 35 envelope. The country panel UI breaks every non-zero contributing signal into a clickable row so users can see exactly which signals moved the score and by how much.

Worked example: Egypt

Egypt's structural components at time of writing: Import Dependency 85, Supplier Concentration 72, Production Trend 55, Food Inflation 38, Climate 58, Conflict 42, Supply-Chain Exposure ≈ 30 . Applying the formula:

```
0.28 · 85 + 0.18 · 72 + 0.14 · 55 + 0.14 · 38 + 0.09 · 58 + 0.09 · 42 + 0.08 · 30  
= 23.80 + 12.96 + 7.70 + 5.32 + 5.22 + 3.78 + 2.40  
= 61.18 → rounds to 61 structural
```

Headline FDRS in Egypt's country panel reads **64** — the 61 structural plus +3 from the nowcast layer (driven by recent FFPI moves and modest IPC pressure on regional neighbours). The country panel exposes both numbers and the +3 breakdown is one click away.

Risk tier classification

0 – 25	Resilient · diversified supply, strong production trend, low live stress
26 – 50	Exposed · structurally vulnerable in 1–2 components but not acutely stressed
51 – 75	Dependent · concentrated import reliance or persistent live signals
76 – 88	Vulnerable · multiple components in the top quartile globally
89 – 100	Severe · system in active crisis (Yemen, S. Sudan, Somalia, Palestine class)

Full data-source inventory

All 33 pipelines are listed below with current health status, what each one feeds downstream, and the provider. Status reflects the `source_manifest.json` snapshot at time of brief generation. Live status fluctuates as upstreams change schemas or rate-limit; the Data Status page on the site is the canonical real-time view.

Source	Status	What it feeds	Provider
WFP HungerMap LIVE	Degraded	Consumption flags, IPC overlay, alerts	WFP
WFP HungerMap per-country	Degraded	FX shock, food inflation, child nutrition	WFP
IPC Acute Food Insecurity	Degraded	Phase classifications (Phase 3+ = crisis)	IPC / WFP mirror
ReliefWeb v2	Live	Humanitarian alerts (Food & Nutrition theme)	OCHA
ACLED	Setup	Conflict events + fatalities (live spike detection)	ACLED
FEWS NET FDW (IPC Outlook)	Live	IPC-style current + 3-month + 8-month phase classifications	USAID / FEWS NET
FAO Food Price Index	Live	Global price-shock signal (5 sub-indices)	FAO
World Bank Pink Sheet	Live	60-month commodity benchmark history	World Bank
UN Comtrade Plus	Live	Bilateral staples trade, 10 HS codes	UN
FAOSTAT TCL net food trade	Degraded	Observed net agri-food trade balance (USD)	FAO
USDA PSD (staples)	Live	Production / imports / exports for wheat, rice, corn, soybean	USDA FAS
FAOSTAT Food Balance Sheets	Degraded	Caloric shares of wheat / rice / maize per country	FAO
World Bank WDI (single)	Live	Most-recent-value per indicator, 249 economies	World Bank
World Bank WDI bulk	Live	20 structural indicators time series	World Bank
World Bank WGI	Degraded	6 governance indicators (rule of law, corruption, ...)	World Bank
World Bank WFSO	Live	Food-security outlook with downside scenarios	World Bank
World Bank LPI	Live	Logistics Performance Index (6 dimensions)	World Bank
World Bank CCKP	Degraded	Climate Change Knowledge Portal — projected ΔT	World Bank
UNDP HDI	Live	Human Development Index 2025 release	UNDP
ND-GAIN Country Index	Degraded	Climate vulnerability + readiness, food sub-score	Notre Dame
WRI Aqueduct 4.0	Degraded	Water stress, drought, riverine + coastal flood risk	WRI
EU JRC INFORM Risk Index	Degraded	Composite humanitarian risk + 3 dimensions	EU JRC
Eurostat food HICP	Live	EU 27 monthly food inflation (CP011)	Eurostat
FAOSTAT food CPI	Degraded	Global annual food consumer-price index	FAO
Open-Meteo Weather	Live	Precipitation + temperature anomalies	Open-Meteo
Open-Meteo Flood	Live	River discharge in flood-exposed nations	Open-Meteo
USGS Water Services	Live	US river-flow anomalies (drought / flood)	USGS
NASA FIRMS VIIRS NRT	Degraded	Active fire detection over cropland	NASA

Source	Status	What it feeds	Provider
OpenAQ v3	Degraded	PM2.5 background — labour + crop stress	OpenAQ
Feeding America (Map the Meal Gap)	Manual	US state food-insecurity %	Feeding America
FAO GIEWS Country Briefs	Manual	Deep-link helper to country briefs	FAO
WFP Annual Country Reports	Manual	Deep-link helper to ACRs	WFP
USDA GAIN Attaché Reports	Manual	Deep-link helper to GAIN reports	USDA FAS

33 pipelines total · 20 healthy · 4 manual · 8 degraded · 1 setup-required · 0 failed. Definition: **healthy** = live or manual. **Degraded** means the script ran without error but the payload is empty, schema-changed, or partial coverage — the downstream calculation falls back to heritage values with the affected component flagged in the UI.

Credential-gated pipelines

Several pipelines required individual application, email correspondence, or registration. These are the ones that took the longest to integrate:

ReliefWeb v2	Approved appname requested by email from apidoc@reliefweb.int. App name = vreeken-foodshie
ACLED	Registered at developer.acleddata.com. Requires both API_KEY and EMAIL secrets.
FEWS NET FDW	Account requested from FEWS NET Help Desk service desk. Script POSTs username/password t
UN Comtrade Plus	Registered at comtradeplus.un.org, subscribed to the API product. Single API_KEY.
OpenAQ	Registered at api.openaq.org/register.
NASA FIRMS	MAP_KEY registered at firms.modaps.eosdis.nasa.gov/api/.

Key design decisions

Two FDRS values, not one

Aid agencies need structural to plan; commodity desks need nowcast to react. A country can be structurally resilient but acutely stressed (a Pakistan flood), or structurally fragile but temporarily stable (a quiet quarter in Yemen). Collapsing both into one number would obscure the difference that actually matters to each audience.

Channel-overlap damper on the stress test

When two or more scenario shocks hit the same FDRS component (e.g. simultaneous wheat and rice export bans both stressing Supplier Concentration), the engine applies $\sqrt{\sum \text{kick}^2}$ rather than straight summation. Without this damper, multi-shock scenarios pegged ~29 countries at 100 simultaneously, which both saturated the colour scale and overstated impact. The damper is documented inline so users can reason about it.

Provenance-aware shocks

Shocks applied to sourced data carry full impact ($\times 1.0$). Shocks applied to legacy curated data are damped to $\times 0.6$. Rationale: a model is only as good as its inputs, and projecting a 20% commodity shock onto a country whose dependency value is hand-curated rather than Comtrade-observed should not produce the same magnitude of FDRS movement as it does for a country with sourced dependency data.

Removing confidence intervals from the UI

An earlier version showed $\pm$ bands and confidence percentages next to FDRS values. These looked like calibrated forecasting statistics but were heuristic placeholders. They were removed and replaced with the explicit note: *'Confidence intervals require a backtest pipeline we haven't built yet — see methodology.'* Honesty about model limits is more useful than fake precision.

Renaming 'Forecast' to 'Modeled Outlook'

The 2030 view used to be labelled 'Forecast', which implied decision-grade probabilistic prediction. It was renamed 'Modeled Outlook' across 21 surfaces. The country chip under the 2030 number reads CURATED baseline + MODELED live overlay, and the methodology explicitly disclaims it as *illustrative scenario estimate, NOT an official FAO-OECD or CCKP forecast*.

Renaming 'Russia halts wheat exports'

The flagship scenario was originally country-named. The maths underneath uses generic supplier-concentration shock, not Russia-specific bilateral simulation. The scenario was renamed to 'Major wheat supplier export shock' with the subtitle: *'Uses supplier concentration exposure; not a Russia-specific bilateral simulation unless Russia-specific dependency data is available.'* Same engine, honest framing.

Removing 'Famine' miscount

An earlier metric displayed 'Famine events: 67'. The underlying count was IPC Phase 3+ areas, not Phase 5 famines. The metric was renamed to 'Hunger alerts' with the bucket key unchanged internally. The point is small but illustrative — the site uses precise IPC terminology because Phase 3 (crisis) and Phase 5 (famine) are categorically different thresholds.

Limitations and honest disclaimers

FoodShield AI is a research artefact, not a calibrated decision-support system. The following limitations are documented inline on the site and should be understood by anyone interpreting the scores.

No back-test pipeline yet

There is no validation that FDRS reliably ranks countries against historical food-security outcomes. Building a back-test pipeline — comparing modelled FDRS in month T against observed IPC phase changes in month T+1, T+3, T+12 — is on the post-launch roadmap. Until then, the score should be read as a structured exposure indicator rather than a predictive probability.

Several climate inputs are currently degraded

CCKP and ND-GAIN have empty or partial coverage in the served snapshots at time of writing. The Climate Vulnerability component (9%) falls back to heritage curated values for affected countries. The 2030 modelled outlook climate kick similarly falls back. Both are flagged in the UI with the explicit 'illustrative' label.

Company sourcing footprints are illustrative

No major agri-food trader publishes per-country sourcing volumes. The Companies tab cards show a directional risk-screen built from reasoned mapping of public reports, not audited data. They are explicitly labelled MODELED and the tab description states this upfront.

Trade-route fallbacks for thin commodities

For the 10 staple HS codes integrated into Comtrade, observed bilateral flows drive the Trade Flow Atlas. For the other 17 commodities (and the 'wide-distribution' cases like wine, dairy, fish, fruit), the dashboard uses a curated COMMODITY_TRADE_ROUTES constant with 436 supplier-mix combinations. Modelled routes are badged MODELED; observed routes are badged OBS · COMTRADE. The distinction is visible everywhere the routes are surfaced.

Coverage gap: per-commodity export destinations

Country export destinations are currently a single flat list per country, not differentiated per commodity. This means export-side visualisations are coarser than import-side ones (which are commodity-specific). Improving this requires a Comtrade pull for the full importer × commodity matrix — on the backlog.

FAOSTAT degradation is structural

FAOSTAT is mid-migration from a legacy guest-token API to a newer authenticated endpoint. The legacy guest flow now returns 403 intermittently. The pipeline is treated as *degradable*: it can fail without breaking the workflow, the source manifest marks it degraded, and the frontend falls back to WFP / Eurostat / World Bank as fresher alternatives. Restoring fully-live FAOSTAT is a separate integration task once the new endpoint stabilises.

Single-file frontend will need splitting

index.html is ~22,750 lines and growing. It's still maintainable today but is approaching the limit where editing locality breaks down. Splitting into per-tab modules with a lightweight bundler is on the post-launch backlog. The choice to launch on a monolith rather than refactor first was deliberate: shipping the working tool to users beat polishing the build pipeline.

Contact and feedback

This brief exists because the project was built solo and would be sharper for outside scrutiny. The kinds of feedback most useful right now:

- • Methodology: are any of the seven components combining things that shouldn't be combined? Is the 28/18/14/14/9/9/8 weighting defensible?
- • Data: which public source is missing from the 33 that would meaningfully sharpen a component?
- • Use cases: would a researcher, journalist, or policy team actually open this twice? Where would they bounce?
- • Honesty: anything on the site that overclaims relative to what the underlying data supports?

Author	Fedde Vreeken
Affiliation	International Economics & Business Economics, Erasmus School of Economics, Rotterdam
Email	fedde.vreeken@gmail.com
LinkedIn	linkedin.com/in/feddevreeken
Site	https://foodshield-ai-fv.vercel.app
Source	https://github.com/feddevreeken-ctrl/foodshield-ai
License	Code MIT · Data attributed to original providers
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FoodShield AI is built on open public data from FAO, the World Bank, USDA, the UN, ACLED, WFP, IPC, FEWS NET, NASA, USGS, the European Commission, OCHA, WRI, the Notre Dame Global Adaptation Initiative, Open-Meteo, OpenAQ, UNDP, Eurostat, and Feeding America. The dashboard is a synthesis layer over their work and is not affiliated with any of them. Where this brief or the site characterises an indicator, the characterisation is the author's own.